

Chapter 27 — Failure Containment

Field	Value
Status	Source-aligned; architecture audit passed
Version	0.1
Scope	Prevent component, node, provider, data, or reasoning failures from cascading across HAL or silently changing its rules.
Constitutional basis	Decisions 19, 22, 28–29, 35, 38, 41–42, 47, 50, 56
Owner review	None required

1. Purpose

Prevent component, node, provider, data, or reasoning failures from cascading across HAL or silently changing its rules.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Failure Domain Manager	Failure boundaries, dependency graph, blast-radius policy, and isolation state
Circuit Breaker Service	Health thresholds, trip state, probes, cooldown, and recovery evidence
Quarantine Manager	Identity, node, provider, artifact, data, or evidence quarantine lifecycle
Degradation Controller	Declared fallback, reduced capability, disclosure, and protected capacity

3. Core State and Records

Object	Required content
Failure Domain	Components, dependencies, protected

Object	Required content
	boundaries, containment action, and safe-state behavior.
Failure Declaration	Trigger, evidence, confidence, affected capability, user impact, and disposition.
Quarantine Record	Subject, reason, restrictions, evidence, authority, review, and release requirements.
Degradation Mode	Unavailable/reduced capability, permitted substitutes, quality impact, disclosure, and exit gate.

4. Runtime Workflow

1. Detect failure from independent health, transaction, evidence, security, or outcome observations.
2. Classify affected domain, consequence, uncertainty, and declared fail-open/fail-closed/fail-safe behavior.
3. Trip the narrowest circuit, isolate dependencies, and preserve constitutional/observability/recovery reserve.
4. Quarantine untrusted identities, nodes, providers, artifacts, or data when integrity is uncertain.
5. Select only predeclared, policy-valid fallback behavior and disclose material degradation.
6. Verify recovery with probes, shadow/canary evidence, and state reconciliation.
7. Restore gradually; protected quarantine release follows its Owner-governed rule.

5. Interfaces and Contracts

Commands, queries, events, records, and evidence are typed, versioned, authenticated, authorized, correlated, and classified. Mutation is accepted only by the authoritative owner. Every external or reality-changing boundary carries identity, policy, authority, provenance, integrity, time, and transaction context.

6. Failure and Recovery

- Failure detector disagreement: preserve both observations and choose the safer bounded state pending evidence.
- Fallback also fails: stop the affected capability rather than cascade through unverified substitutes.
- Containment harms unrelated work: record impact and adjust topology without lowering constitutional controls.
- Unknown root cause: retain quarantine and continue evidence collection; uncertainty is not restoration evidence.

7. Constitutional Guarantees

- No single failure may silently change HAL's rules or constitutional behavior.
- Failures are contained by explicit domains rather than implicit shared dependencies.
- Graceful degradation is transparent, bounded, and policy-governed.
- Quarantine limits authority and access without rewriting identity or historical evidence.

8. Prohibited Behaviors

- No component may silently broaden or narrow constitutional authority, reinterpret Owner intent, erase dissent, or promote derived state to authority.
- No failure, degraded dependency, emergency, or optimization may silently change identity, policy, trust, authentication, evidence, or verification rules.
- No model, service, provider, node, Presence, credential, relationship, environment, or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records identity, correlation, authority/policy context, inputs, state versions, decisions, limitations, failures, recovery, and outcomes. Verification is risk-scaled, reproducible, and stored as durable evidence. Status and explanation derive from governed state rather than model assertion.

10. Source Alignment and Review

This chapter implements Decisions 19, 22, 28–29, 35, 38, 41–42, 47, 50, 56. Final source-alignment and constitutional disposition is recorded in the batch audit reports; embedded approval text is not treated as review evidence.